



University of Asia Pacific

Admit Card

Mid-Term Examination of Fall, 2024

Financial Clearance

PAID

Registration No : 22201243

Student Name : H. M. Tahsin Sheikh

Program : Bachelor of Science in Computer Science and Engineering



SI.NO.	COURSE CODE	COURSE TITLE	CR.HR.	EXAM. SCHEDULE
1	CSE 301	Object Oriented Programming II and Web Programming	3.00	
2	CSE 302	Object Oriented Programming II and Web Programming Lab	1.50	
3	CSE 303	Computer Architecture and Organization	3.00	
4	CSE 305	System Analysis and Design	3.00	
5	CSE 307	Project Strategy and Management	0.00	
6	BUS 301	Business and Entrepreneurship	3.00	
7	ENG 300	Technical Writing and Presentation	1.50	
8	BIE 301	Bioinformatics Engineering	3.00	

Total Credit: 18.00

1. Examinees are not allowed to enter the examination hall after 30 minutes of commencement of examination for mid semester examinations and 60 minutes for semester final examinations.
2. No examinees shall be allowed to submit their answer scripts before 50% of the allocated time of examination has elapsed.
3. No examinees would be allowed to go to washroom within the first 60 minutes of final examinations.
4. No student will be allowed to carry any books, bags, extra paper or cellular phone or objectionable items/incriminating paper in the examination hall. Violators will be subjects to disciplinary action.

This is a system generated Admit Card. No signature is required.

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Fall 24
Program: B.Sc. in CSE
3rd year 1st Semester

Course Title: Systems Analysis and Design

Time: 1 hour.

Credit Hour: 3.00

Course Code: CSE 305

Full Marks: 20

There are **Three** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [5 MARKS]

- a. Identify the basic principles of Agile modeling. [5]

QUESTION 2 [10 MARKS]

- a. Activities A to K are required to complete a project. The expected time and the immediate predecessors of these activities are given in the table. Calculate the overall completion period of the project considering the slack time. [10]

Activity	Time (hours)	Immediate predecessors
A	2	—
B	3	—
C	2	—
D	4	A
E	5	B
F	4	B
G	3	C
H	10	D, E
I	5	F
J	8	G
K	3	H, I, J

QUESTION 3 [5 MARKS]

- a. Explain a context level Data Flow Diagram (DFD 0) for online shopping system. [5]

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Fall 24
Program: B.Sc. in CSE
3rd year 1st Semester

Course Title: Object Oriented Programming II and Web Programming
Time: 1 hour.

Credit Hour: 3.00

Course Code: CSE 301
Full Marks: 20

There are **Three** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. You are tasked with implementing a Python function named 'convert_to_fahrenheit' to accomplish this conversion for a list of temperature readings. The function should take a list of temperature readings as input in celsius and return a list containing the fahrenheit temperature for each temperature reading. [5]

To convert temperatures from Celsius to Fahrenheit, you can use the following formula:

$$Fahrenheit = (9 \div 5) \times Celsius + 32$$

Sample Input	Sample Output
celsius_temp = [0, 25, -10, 100]	[32.0, 77.0, 14.0, 212.0]

- b. Write a Python program that takes a sentence as input and counts the occurrences of each vowel (a, e, i, o, u). Store the counts in a dictionary where the vowels are the keys and their respective counts are the values. [5]

Sample input	Sample output
Enter a sentence: This is a sample sentence.	{ 'a': 2, 'e': 4, 'i': 2, 'o': 0, 'u': 0 }

QUESTION 2 [5 MARKS]

a. Eid Party Guest Management - OOP Task:

[5]

Create a class **Guest** with attributes for **name**, **age**, and **favorite dish**. Add a method **show_details()** to display the guest's information and a method **eid_greeting()** to print "Eid Mubarak, [name]!" without taking any extra arguments. Create at least two guest objects and call both methods for each guest.

If a guest named **Aisha**, aged **25**, loves **Biryani**, the output should be:

when **show_details()** is called:

Guest Name: Aisha

Age: 25

Favorite Dish: Biryani

When **eid_greeting()** is called:

Eid Mubarak, Aisha!

QUESTION 3 [5 MARKS]

a. Write code to create an HTML form with the following elements:

[5]

- An input field to enter the user's name.
- A dropdown list to select the user's favorite food with options like Pizza, Pasta, Burger, Salad.
- A radio button with options Vegetarian or Non-Vegetarian (only one option should be selectable).
- A checkbox to confirm if the user has any food allergies.
- A submit button to submit the form

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Fall 2024
Program: B.Sc. in CSE
3rd year 1st Semester

Course Title: Computer Architecture and Organization
Time: 1 hour.

Credit Hour: 3.00

Course Code: CSE 303
Full Marks: 20

There are **Two** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. Draw the basic components and layers of a computer. Show the relationship among instruction set, software, and hardware that define computer architecture. [2]
- b. Illustrate the 1st version Algorithm step by step to multiply: 5 x 4 using 4bits multiplier. [4]
- c. A compiler designer tries to decide between two code sequences for a particular computer. The hardware designers have supplied the following facts: [4]

Table 1. CPI for this instruction class

CPI	CPI for this instruction class		
	A	B	C
	2	3	1

For a particular high-level language statement, the compiler writer is considering two code sequences that require the following instruction counts:

Table 2. Instruction count for instruction class

Code sequence	Instruction count for instruction class		
	A	B	C
1	3	3	4
2	4	3	5

Which code sequence executes the most instructions? Which will be faster?
QUESTION 2 [10 MARKS]

a) According to the MIPS instruction set Architecture, answer the following: [7]

for this high-level statement, $A = X[i+20] - X[i+10] + 40$;

where i is the last digit of your registration number.

Illustrate the MIPS assembly code and MIPS machine code.

[3]

b) Identify the following MIPS instructions and write the corresponding MIPS assembly code.

0000001000110010100000000100001
10

N.B : Add = 32; addi = 33; Subi = 22; lw = 42, sw = 35, sub = 34, s0 = 15, s1 = 16, s2 = 17, s3 = 18, to = 8;

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Mid Semester Examination, Fall 24
Program: B.Sc. in CSE
3rd year 1st Semester

Course Title: Bioinformatics Engineering
Time: 1 hour. Credit Hour: 3.00

Course Code: BIE 301
Full Marks: 20

There are Two Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [8 MARKS]

- a. Define i. Genotype ii. Phenotype iii. Dominant Gene iv. Transcription [4]
b. Explain the Greedy Reversal algorithm for genome rearrangement with an example. [4]

QUESTION 2 [12 MARKS]

- a. Find the optimal global alignment of the following two sequences using -2 as the gap penalty, -1 as the mismatch penalty, and 2 as the score for a match. Show the corresponding dynamic programming (DP) table. Please mark the paths, which correspond to the optimal alignments, in the DP table. [5]

Sequence 1 = GTAGCTA

Sequence 2 = ACGTGCTAGC

- b. Following shows three alignments of five sequences obtained from different algorithms. [7]
Determine the best alignment based on entropy.

Alignment 1:

ACGTACGTAA
TCGAACGTAT
GGGTACGGGA
CCGTACGTCA
TAATTCGTG-

Alignment 2:

ACGTACG-TAA
TCGAACGT-AT
GGGTACGG-GA
CCGTACG-TCA
TAATTCGT-GA

Alignment 3:

ACGTA-CG-TAA
TCGAA-CGT-AT
GGGT-ACGGGA
CCGTACGT-CA
TAATTC-GTGA

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Fall 24
Program: B.Sc. in CSE
3rd year 1st Semester

Course Title: Business and Entrepreneurship
Time: 1 hour

Credit Hour: 3.00

Course Code: BUS 301
Full Marks: 20

Instructions:

1. Each question has two parts. Answer one or the other.
2. Ensure clarity and coherence in your responses.

QUESTION 1 [Marks 10]

- a. Define Business in your own words. Explain the elements of the Business Environment with at least 2 examples. [1+2]

OR

Briefly compare a startup and an SME in terms of business model and funding approach. Provide one example for each.

- b. List the characteristics of successful entrepreneurs. Explain the criticism: "9 out of 10 businesses fail in their first few years." [1+2]

OR

How does market research help entrepreneurs reduce failure risks? Provide one real-world example.

- c. Write down the types of companies in Bangladesh as per the Companies Act, 1994. Explain the steps for the formation of a private limited company. [1+3]

OR

Identify two key types of innovation in entrepreneurship and briefly explain how they impact business success.

QUESTION 2 [Marks 4]

A company is considering two projects, X and Y. The expected returns from these projects are as follows. Calculate the **Standard Deviation and Coefficient of variation** and select the less risky project. [4]

Year	Project A %	Project B %
1	10%	15%
2	8%	13%
3	13%	18%
4	12%	17%

OR

Case Study: A startup is launching an online service in a competitive market. Briefly outline a simple Go-To-Market (GTM) strategy, focusing on customer segmentation and pricing.

QUESTION 3 [Marks 6]

How does design thinking help in developing customer-centric products? Give one real-world example. Explain the "Problem-Solution Fit" approach in business. How does it help in startup success? 6

OR

A company is considering three projects (A, B, and C) that are mutually inclusive.

Year	Project A BDT	Project B BDT	Project C BDT
0 (Initial Investment)	-120,000	-180,000	-150,000
1	50,000	60,000	70,000
2	50,000	65,000	75,000
3	50,000	70,000	80,000
4	50,000	80,000	90,000

Conditions & Constraints:

1. The total budget for selected projects must not exceed BDT 300,000.
2. The company wants to select projects based on the NPV while maintaining the Payback Period (if NPVs are similar).
3. The discount rate is 10%.

Calculate the **Payback Period** and NPV for each project individually. Determine which project/projects should be selected based on the budget constraint. If the company finds a way to increase its budget to \$330,000, would the decision change? Why or why not? 6